

NCERT Exemplar Solutions for Class 12

Chemistry

Chapter 8 - The d- and f- Block Elements

Exercise

Multiple Choice Question (Type-I)

1. Electronic configuration of a transition element X in +3 oxidation state is $[\text{Ar}] 3d^5$. What is its atomic number?

(A) 25

(B) 26

(D) 27

(D) 24

Ans: Positive oxidation states indicate the loss of electrons from the atom. If X is in +3 oxidation state, then three electrons have been removed from it. Find the atomic number of parent atom, we will add three in the given electronic number. i.e.

$$X^{3+} (Z = 23) = [\text{Ar}] 3d^5$$

$$X = 23 + 3 = 26$$

Correct answer: (B)

2. The electronic configuration of Cu(II) is $3d^9$ whereas that of Cu(I) is $3d^{10}$. Which of the following is correct?

(a) Cu(II) is more stable .

(b) Cu(II) is less stable

(c) Cu(I) and Cu(II) are equally stable

(d) Stability of Cu(II) and Cu(I) depends on nature of copper salts.

Ans: Electronic configuration of Cu is $[\text{Ar}] 3d^{10} 4s^1$

Cu(I) - $[\text{Ar}] 3d^{10} 4s^0$

Cu(II) - $[\text{Ar}] 3d^9 4s^0$

Despite the fact that Cu(I) has fully filled 3d-orbital but Cu(II) is more stable than Cu(I) due to the greater effective nuclear charge of Cu(II) as nucleus has to hold 17 electrons rather than 18 electrons like in Cu(I).

Correct answer: (A)

3. Metallic radii of some transition elements are given below. Which of these elements will have highest density?

Element	Fe	Co	Ni	Cu
Metallic radii/pm	126	125	125	128

(A) Fe

(B) Ni

(C) Co

(D) Cu

Ans: Density is mass by volume. As we move from left to right a long period, the atomic radii decreases. Hence, volume decreases. Also increase in atomic masses is observed.

So overall the density increases out of above option from iron to copper, copper will have highest density.

Correct answer: (D)

4. Generally, transition elements form coloured salts due to the presence of unpaired electrons. Which of the following compound will be coloured in solid state?

- (A) Ag_2SO_4
- (B) CuF_2
- (C) ZnF_2
- (D) Cu_2Cl_2

Ans: (A) Ag_2SO_4 (Ag^+) $\rightarrow 5d^{10} 6s^0$

- (B) CuF_2 (Cu^{2+}) $\rightarrow 3d^9 4s^0$
- (C) ZnF_2 (Zn^{2+}) $\rightarrow 3d^{10} 4s^0$
- (D) Cu_2Cl_2 (Cu^+) $\rightarrow 3d^{10} 6s^0$

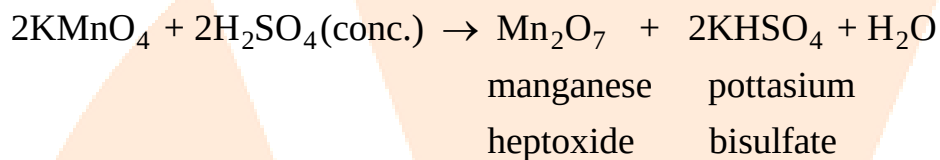
Unpaired electrons present in any compound impart colour to the salt of transition metal. Only CuF_2 has unpaired electrons in its 3d orbital that's why it is white coloured in its solid-state while rest of the salts are colourless.

Correct answer: (B)

5. On addition of small amount of KMnO_4 to concentrated H_2SO_4 , a green oily compound is obtained which is highly explosive in nature. Identify the compound from the following:

- (A) Mn_2O_7
- (B) MnO_2
- (C) MnSO_4
- (D) Mn_2O_3

Ans:



Manganese heptoxide is an acid anhydride of permanganic acid (HMnO_4). It is a dangerous oxidiser, volatile and highly reactive liquid.

Correct answer: (A)

6. The magnetic nature of elements depends on the presence of unpaired electrons. Identify the configuration of transition element, which shows the highest magnetic moment.

- (A) $3d^7$
- (B) $3d^5$
- (C) $3d^8$
- (D) $3d^2$

Ans: Magnetic moment of an electron/dipole moment is caused by its intrinsic properties of spin and electric charge. It depends upon the number of unpaired electrons in its valance shell. The more the number of unpaired electrons, the greater will be the value of magnetic moment.

$3d^7 = 3$ unpaired electrons

$3d^5 = 5$ unpaired electrons

$3d^8 = 2$ unpaired electrons

$3d^2 = 2$ unpaired electrons

Out of all, $3d^5$ has five unpaired electrons that is maximum and hence it has the highest magnetic moment.

Correct answer: (B)

7. Which of the following oxidation state is common for all lanthanides?

(A) +2

(B) +3

(C) +4

(D) +5

Ans: Out of all +3 is the most common oxidation State for all the lanthanide elements.

Correct answer: (B)

8. Which of the following reactions are disproportionation reactions?

(A) $\text{Cu}^+ \rightarrow \text{Cu}^{2+} + \text{Cu}$

(B) $3\text{MnO}_4^{4-} + 4\text{H}^+ \rightarrow 2\text{MnO}_4^- + \text{MnO}_2 + 2\text{H}_2\text{O}$

(C) $2\text{KMnO}_4 \rightarrow \text{K}_2\text{MnO}_4 + \text{MnO}_2 + \text{O}_2$

(D) $2\text{MnO}_4^- + 3\text{Mn}^{2+} + 2\text{H}_2\text{O} \rightarrow 5\text{MnO}_2 + 4\text{H}^+$

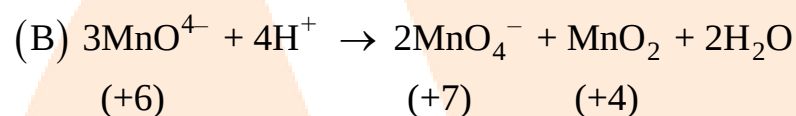
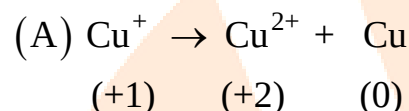
(i) a, b

(ii) a, b, c

(iii) b, c, d

(iv) a, d

Ans: Disproportionation/redox reaction is a reaction in which one compound is oxidised as well as reduced at the same time.



Correct answer: (i)

9. When KMnO_4 solution is added to oxalic acid solution, the decolourisation is slow in the beginning but becomes instantaneous after some time because

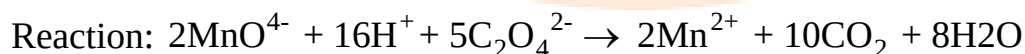
(A) CO_2 is formed as the product

(B) Reaction is exothermic

(C) MnO_4 catalyses the reaction

(D) Mn^{2+} acts as auto catalyst

Ans: Mn^{2+} formed in the reaction acts as an auto catalyst. Auto catalyst is a compound which is one of the products that are formed in the reaction itself catalyses the reaction.



Correct answer: (D)

10. There are 14 elements in actinide series. Which of the following element does-not belong to this series?

- (A) U
- (B) Np
- (C) Tm
- (D) Fm

Ans: In the periodic table, actinides lie from atomic number 90 to 103.

U - $Z = 92$

Np - $Z = 93$

Tm - $Z = 69$

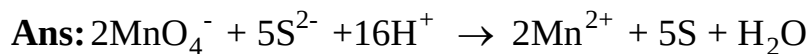
Fm - $Z = 100$

Thulium is not an actinide; it is a lanthanide.

Correct answer: (C)

11. KMnO_4 acts as an oxidizing agent in acidic medium. The number of moles of KMnO_4 that will be needed to react with one mole of sulphide ions in acidic solution is

- (A) $\frac{2}{5}$
- (B) $\frac{3}{5}$
- (C) $\frac{4}{5}$
- (D) $\frac{1}{5}$



From the reaction,

For 5 moles of S , two moles of KMnO_4 were required.

Therefore, for one mole of S , the number of moles of KMnO_4 required will be = $\frac{2}{5}$.

Correct answer: (A)

12. Which of the following is amphoteric oxide?

$\text{Mn}_2\text{O}_7, \text{CrO}_3, \text{Cr}_2\text{O}_3, \text{CrO}, \text{V}_2\text{O}_5, \text{V}_2\text{O}_4$

- (A) $\text{V}_2\text{O}_5, \text{Cr}_2\text{O}_3$
- (B) $\text{Mn}_2\text{O}_7, \text{CrO}_3$
- (C) $\text{CrO}, \text{V}_2\text{O}_5$
- (D) $\text{V}_2\text{O}_5, \text{V}_2\text{O}_4$

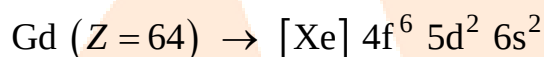
Ans: Oxides in the lower oxidation state are basic and in the higher oxidation state they are acidic. They are amphoteric in the intermediate oxidation state. Therefore, $\text{V}_2\text{O}_5, \text{Cr}_2\text{O}_3$ are amphoteric in nature.

Correct answer: (A)

13. Gadolinium belongs to 4f series. Its atomic number is 64. Which of the following is the correct electronic configuration of gadolinium?

- (A) $[\text{Xe}] 4f^7 5d^1 6s^2$
- (B) $[\text{Xe}] 4f^6 5d^2 6s^2$
- (C) $[\text{Xe}] 4f^8 6d^2$
- (D) $[\text{Xe}] 4f^9 5s^1$

Ans: Gadolinium belongs to 4f series it has atomic no.= 64. It has extra stability due to half-filled 4f sub shell.



Correct answer: (A)

14. Interstitial compounds are formed when small atoms are trapped inside the crystal lattice of metals. Which of the following is not the characteristic property of interstitial compounds?

- (A) They have high melting points in comparison to pure metals
- (B) They are very hard
- (C) They retain metallic conductivity
- (D) They are chemically very reactive.

Ans: Interstitial compounds/alloys are substances that are formed when a small atom like carbon, hydrogen, boron, nitrogen occupy can spaces in their lattices. All the properties mentioned above are true for interstitial compounds except (D), they are chemically inert.

Correct answer: (D)

15. The magnetic moment is associated with its spin angular momentum and orbital angular momentum. Spin only magnetic moment value of Cr^{3+} ion is

(A) 2.87 B.M.

(B) 3.87 B.M.

(C) 3.47 B.M.

(D) 3.57 B.M.

Ans: Cr $\rightarrow$ (Z=24)

Cr³⁺ $\rightarrow$ (Z=21)

1s² 2s² 2p⁶ 3s² 3p⁶ 3d³ 4s⁰

n = 3 (3 unpaired electrons)

$$\mu = \sqrt{n(n+2)}$$

$$\mu = \sqrt{3(3+2)}$$

$$\mu = \sqrt{3(5)}$$

$$\mu = \sqrt{15} = 3.87 \text{ B.M.}$$

Correct answer: (B)

16. KMnO₄ acts as an oxidizing agent in alkaline medium. When alkaline KMnO₄ is treated with KI, iodide ion is oxidized to

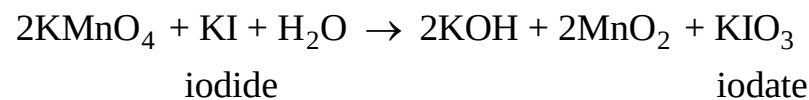
(A) I₂

(B) IO⁻

(C) IO₃⁻

(D) IO₄⁻

Ans: Iodide is oxidised to iodate.



Correct answer: (C) IO_3^-

17. Which of the following statement is not correct?

(A) Copper liberates hydrogen from acids

(B) In its higher oxidation states, manganese forms stable compounds with oxygen and fluorine

(C) Mn^{3+} and Co^{3+} are oxidizing agents in aqueous solution

(D) Ti^{2+} and Cr^{2+} are reducing agents in aqueous solution

Ans: Copper lies below hydrogen in electrochemical series and hence cannot liberate hydrogen from acids. (A) is incorrect.



Correct answer: (A)

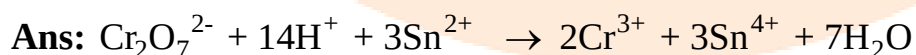
18. When acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution is added to Sn^{2+} salts, Sn^{2+} changes to

(A) Sn

(B) Sn^{3+}

(C) Sn^{4+}

(D) Sn^+



In the above redox reaction, oxidation of Sn^{2+} is taking place. It gets converted to Sn^{4+} whereas, reduction of chromium is occurring from +6 oxidation state to +3.

Correct answer: (C)

19. Highest oxidation state of manganese in fluoride is +4 (MnF_4) but highest oxidation state in oxides is +7 (Mn_2O_7) because

- (A) fluorine is more electronegative than oxygen**
- (B) fluorine does not possess d-orbitals**
- (C) fluorine stabilizes lower oxidation state**
- (D) in covalent compounds fluorine can form single bond only while oxygen forms double bond**

Ans: Oxygen has the capacity to form multiple bonds which enables it to form a variety of covalent compounds.

In (Mn_2O_7) also, 6 oxygen are doubly bonded to two manganese atoms and one oxygen is forming bridge between two.

While in (MnF_4), four fluorine atoms are singly bonded to manganese atom giving it a +4 oxidation state.

Therefore, due to capability of oxygen to have multiple bonds in covalent compounds, manganese is having higher oxidation state of +7 in (Mn_2O_7) .

Correct answer: (D)

20. Although zirconium belongs to 4d transition series and hafnium to 5d transition series even then they show similar physical and chemical properties because

- (A) both belong to d-block**
- (B) both have same number of electrons**

(C) both have similar atomic radius

(D) both belong to the same group of the periodic table

Ans: As an effect of lanthanide contraction, zirconium and hafnium have similar radius of 160pm and 159 pm respectively. Due to this similarity in their size, they show similar physical and chemical properties.

Lanthanide contraction: Because the elements in Row 3 have 4f electrons. These electrons do not shield good, causing a greater nuclear charge. This greater nuclear charge has a greater pull on the electrons and result in the decrease in their size and atomic radii.

Correct answer: (C)

21. Why is HCl not used to make the medium acidic in oxidation reactions of KMnO_4 in acidic medium?

(A) Both HCl and KMnO_4 act as oxidizing agents.

(B) KMnO_4 oxidises HCl into Cl_2 which is also an oxidizing agent.

(C) KMnO_4 is a weaker oxidizing agent than HCl.

(D) KMnO_4 acts as a reducing agent in the presence of HCl.

Ans: HCl is not used in titrations involving KMnO_4 as it produces unsatisfactory results. When HCl is added to KMnO_4 , it itself acts as reducing agent and converts HCl to Cl_2 .

Correct answer: (B)

Multiple Choice Questions (Type-II)

22. Generally, transition-elements and their salts are coloured due to the presence of unpaired electrons in metal ions. Which of the following compounds are coloured?

- (A) KMnO_4
- (B) $\text{Ce}(\text{SO}_4)_2$
- (C) TiCl_4
- (D) Cu_2Cl_2

Ans: Due to charge transfer phenomenon, which can be understood, for an instant, an electron is transferred from oxygen to the central metal atom, where oxidation state of metal is reduced by one and oxygen gets converted from O^- to O^{2-} .

This transfer imparts Colour to the compound.

Correct answer: (A) & (B)

23. Transition elements show magnetic moment due to spin and orbital motion of electrons. Which of the following metallic ions have almost same spin only magnetic moment?

- (A) Co^{2+}
- (B) Cr^{2+}
- (C) Mn^{2+}
- (D) Cr^{3+}

Ans: Magnetic moment of an electron/dipole moment is caused by its intrinsic properties of spin and electric charge. It depends upon the number of unpaired electrons in its valance shell. The more the number of unpaired electrons, the greater will be the value of magnetic moment.

Co^{2+} and Cr^{3+} have same unpaired electrons i.e. 3 unpaired electrons. Hence they have same spin only magnetic moment.

Correct answer: (A) & (D)

24. In the form of dichromate, Cr (VI) is a strong oxidizing agent in acidic medium but Mo (VI) in MoO_3 and W (VI) in WO_3 are not because

- (a) Cr (VI) is more stable than Mo (VI) and W (VI)**
- (b) Mo (VI) and W (VI) are more stable than Cr (VI)**
- (c) higher oxidation states of heavier members of group-6 of transition series are more stable**
- (d) lower oxidation states of heavier members of group-6 of transition series are more stable.**

Ans: For the heavier metals of d-block, higher oxidation state is more favourably stable than the lower oxidation state. For example: Mo (VI) and W (VI) are more stable than Cr (VI). For lighter d-block elements like chromium, +3 oxidation state is more stable. So, Cr (VI) in dichromate act as a strong oxidising agent by reducing itself from +6 to +3 oxidation states While Mo (VI) and W (VI) does not.

Correct answer: (B) and (C)

25. Which of the following actinoids show oxidation states up to +7?

- (A) Am**
- (B) Pu**
- (C) U**
- (D) Np**

Ans: Actinides show a variety of oxidation states like +3, +5, +6 and +7.

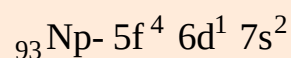
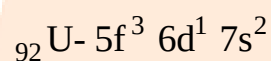
From the given option, plutonium and neptunium show all oxidation states up to +7.

Correct answer: (B) & (D)

26. General electronic configuration of actinoids is $(n-2)f^{1-14}(n-1)d^{0-2}ns^2$. Which of the following actinoids have one electron in 6d orbital?

- (A) U (Atomic no. 92)
- (B) Np (Atomic no. 93)
- (C) Pu (Atomic no. 94)
- (D) Am (Atomic no. 95)

Ans: Electronic configuration:



Correct answer: (A) and (B)

27. Which of the following lanthanoids show +2 oxidation state besides the characteristic oxidation state +3 of lanthanoids?

- (A) Ce
- (B) Eu
- (C) Yb
- (D) Ho

Ans: $(Z=57) = [\text{Xe}] 4f^5 5d^0 6s^2$, O.S. = +3,+4

Eu $(Z=63) = [\text{Xe}] 4f^7 5d^0 6s^2$, O.S. = +2,+3

Yb (Z=70) = [Xe] $4f^{14} 5d^0 6s^2$, O.S. = +2,+3

Ho (Z=67) = [Xe] $4f^{11} 5d^0 6s^2$, O.S. = +3

Correct answer: (B) and (C)

28. Which of the following ions show higher spin only magnetic moment value?

(A) Ti^{3+}

(B) Mn^{2+}

(C) Fe^{2+}

(D) Co^{3+}

Ans: Magnetic moment of an electron/dipole moment is caused by its intrinsic properties of spin and electric charge. It depends upon the number of unpaired electrons in its valance shell. The more the number of unpaired electrons, the greater will be the value of magnetic moment.

Electronic configuration:

$Ti^{3+} = 3d^1 = 1$ unpaired electron

$Mn^{2+} = 3d^5 = 5$ unpaired electrons

$Fe^{2+} = 3d^6 = 4$ unpaired electrons

$Co^{2+} = 3d^7 = 3$ unpaired electrons

Out of all given options, Mn^{2+} and Fe^{2+} have unpaired electrons in number of five and four respectively. Hence, have higher spin only magnetic moment.

Correct answer: (B) and (C)

29. Transition elements form binary compounds with halogens. Which of the following elements will form MF_3 type compounds?

- (A) Cr
- (B) Co
- (D) Cu
- (D) Ni

Ans: Cr and Co form tri halide compounds.

Correct answer: (A) and (B)

30. Which of the following will not act as oxidizing agents?

- (A) CrO_3
- (B) MoO_3
- (C) WO_3
- (D) CrO_4^{2-}

Ans: A compound act as oxidising agent when it's central atom is reduced to its lower oxidation state. This occurs only when the lower oxidation state of metal is more stable than the higher oxidation states.

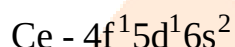
In metal WO_3 and CrO_4^{2-} , both W and Cr are stable in their higher oxidation state and won't get reduced to their lower oxidation state. Therefore, will not act as oxidising agents.

Correct answer: (B) and (C)

31. Although +3 is the characteristic oxidation state for lanthanoids but cerium also shows +4 oxidation state because

- (A) it has variable ionization enthalpy
- (B) it has a tendency to attain noble gas configuration
- (C) it has a tendency to attain f^0 configuration
- (D) it resembles Pb^{4+}

Ans: Electronic configuration of cerium:



Lead has the tendency to lose its four outer electrons to attain Noble gas configuration and this removal of electrons corresponds to the f^0 configuration and cause it to be stable in +4 oxidation state.

Correct answer: (B) and (C)

Short Answer Type Questions

32. Why does copper not replace hydrogen from acids?

Ans: Copper lies below hydrogen in electrochemical series which means it has positive standard reduction potential i.e. $+0.34\text{ V}$. and hence cannot liberate hydrogen from acids.

33. Why E° value for Mn, Ni and Zn are more negative than expected?

Ans: Mn has half-filled $3d^5$ electrons and Zn has fully-filled $3d^{10}$ electrons which give them extra stability and they both resist to lose electrons and get reduced.

The E° value also depend on the hydration enthalpy. More negative is the enthalpy of hydration, high is the E° value. This is the reason behind Ni having higher E° value than expected.

34. Why first ionization enthalpy of Cr is lower than that of Zn ?

Ans: $\text{Cr (24)} = [\text{Ar}] 3d^5 4s^2$

$\text{Zn (30)} = [\text{Ar}] 3d^{10} 4s^2$

The stability of orbitals according to the electrons filled is in order:

Fully filled > half-filled > partially filled

As $3d^{10}$ is completely filled orbital of Zn is more stable than the $3d^5$ half-filled orbital of Cr . It is comparatively easy to remove electron from $3d^5$ than from $3d^{10}$.

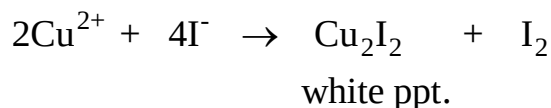
Therefore, first ionization enthalpy of Cr is lower than that of Zn .

35. Transition elements show high melting points. Why?

Ans: Transition elements have ability to involve their d-electrons along with their s-electrons in the process of bond formation. This results in formation of strong metal-metal bond which makes their melting point high.

36. When Cu^{2+} ion is treated with KI, a white precipitate is formed. Explain the reaction with the help of chemical equation.

Ans: The reactions will be as:



When Cu^{2+} reacts with potassium iodide white precipitate of Cu_2I_2 is formed.

37. Out of Cu_2Cl_2 and CuCl_2 , which is more stable and why?

Ans: Higher enthalpy of hydration provides extra stability to Cu^{2+} oxidation state than +1 oxidation state.

Enthalpy of hydration is described as the amount of energy released on dilution of one mole of gaseous ions.

38. When a brown compound of manganese

(A) is treated with HCl , it gives a gas

(B) The gas taken in excess, reacts with NH_3 to give an explosive compound

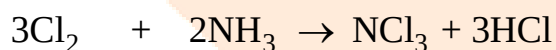
(C) Identify compounds A, B and C.

Ans: The reactions are explained as:



(A)

(B)



(excess)

(C)

39. Although fluorine is more electronegative than oxygen, but the ability of oxygen to stabilize higher oxidation states exceeds that of fluorine. Why?

Ans: Fluorine - $1s^2 2s^2 2p^5$

Oxygen - $1s^2 2s^2 2p^6 3s^2 3p^6$

As we can infer from the electronic configuration, fluorine only has one unpaired electron in its 2p-orbital which can be involved in formation of only one bond and fluorine lacks d-orbital.

While oxygen has d-orbitals in its configuration which allow the oxygen to form multiple bonds. Therefore, oxygen has greater stability to stabilise its oxidation states than fluorine.

40. Although Cr^{3+} and Co^{2+} ions have same number of unpaired electrons but the magnetic moment of Cr^{3+} is 3.87 BM and that of Co^{2+} is 4.87 BM. Why?

Ans: Magnetic moment of an electron/dipole moment is caused by its intrinsic properties of spin and electric charge. It depends upon the number of unpaired electrons in its valance shell. The more the number of unpaired electrons, the greater will be the value of magnetic moment.

$\text{Cr}^{3+} = [\text{Ar}] 3d^3 = 3 \text{ unpaired electrons}$

$\text{Co}^{2+} = [\text{Ar}] 3d^7 = 3 \text{ unpaired electrons}$

But as both the ions have same no. Of unpaired electrons, there's one other factor called orbital contribution which also influences the magnetic moment. As, the three unpaired electronic configuration is symmetrical in Cr^{3+} this contributes zero orbital contribution but this is quite appreciable in case of Co^{2+} and hence it has larger value of Magnetic moment than Cr^{3+}

41. Ionisation enthalpies of Ce, Pr and Nd are higher than Th, Pa and U. Why?

Ans: Ce, Pr and Nd belong to the lanthanoid series whereas Th, Pa and U belong to the actinoids family.

When electrons start accommodating the 4f and 5f orbitals, the 5f electrons penetrate less into the inner core. They are more effectively shielded nucleus as comparison to 4f-electrons in lanthanoids. This leads to the fact that 5f-electrons experience reduced nuclear force of attraction and hence they have Lower ionisation enthalpies than lanthanoids.

42. Although Zr belongs to 4d and Hf belongs to 5d transition series but it is quite difficult to separate them. Why?

Ans: As an effect of lanthanoid contraction, zirconium and hafnium have similar radius of 160pm and 159 pm respectively. Due to this similarity in their size, they show similar physical and chemical properties.

Lanthanoid contraction: Because the elements in Row 3 have 4f electrons. These electrons do not shield good, causing a greater nuclear charge. This greater nuclear charge has a greater pull on the electrons and result in the decrease in their size and atomic radii.

43. Although +3 oxidation state is the characteristic oxidation state of lanthanoids but cerium shows +4 oxidation state also. Why?

Ans: $\text{Ce} = 4f^1 5d^1 6s^2$

$\text{Ce}^{3+} = 4f^1 5d^0 6s^0$

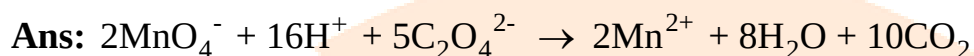
As in Ce^{3+} ,

4f has only single electron. Cerium holds the capacity to lose this electron also and attain $4f^0$ configuration which is more stable.

$\text{Ce}^{4+} = 4f^0 5d^0 6s^0$

Hence it shows 4+ oxidation state.

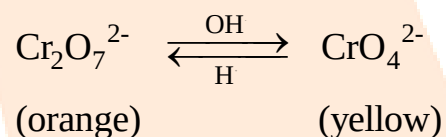
44. Explain why does colour of KMnO_4 disappear when oxalic acid is added to its solution in acidic medium?



KMnO_4 oxidises the oxalic acid to CO_2 and reduces itself to Mn^{2+} state. Mn^{2+} is colourless that's why it seems that KMnO_4 has been disappeared.

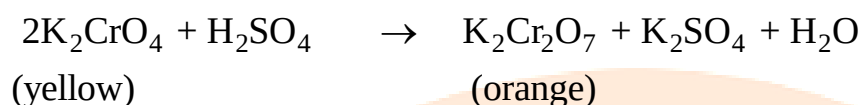
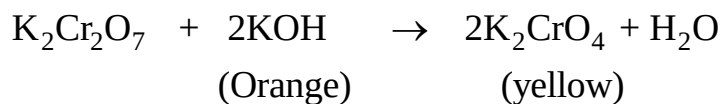
45. When orange solution containing $\text{Cr}_2\text{O}_7^{2-}$ ion is treated with an alkali, a yellow solution is formed and when H^+ ions are added to yellow solution, an orange solution is obtained. Explain why does this happen?

Ans: This is due to the inter conversion of dichromate (orange) to chromate ion (yellow).



46. A solution of KMnO_4 on reduction yields either a colourless solution or a brown precipitate or a green solution depending on pH of the solution. What different stages of the reduction do these represent and how are they carried out?

Ans: KMnO_4 act as an oxidising agent however it's activity as oxidising agent is influenced by the pH of the solution i.e. acidic, basic or neutral solution.



47. The second and third rows of transition elements resemble each other much more than they resemble the first row. Explain why?

Ans: As an effect of lanthanoid contraction, the second and third rows of transition elements resemble each other. For example, zirconium and hafnium have similar radius i.e. 160pm and 159pm respectively. Due to this similarity in their size, they show similar physical and chemical properties.

Lanthanoid contraction: Because the elements in Row 3 have 4f electrons. These electrons do not shield good, causing a greater nuclear charge. This greater nuclear charge has a greater pull on the electrons and result in the decrease in their size and atomic radii.

48. E of Cu is +0.34 V while that of Zn is -0.76 V. Explain.

Ans: The sum of sublimation energy and ionisation enthalpy to oxidise Cu(s) to Cu^{2+} is so highly that it is not compensated by the hydration enthalpy of Cu. Due to this, the E of Cu is positive.

While in case of Zn, the E value is negative or more negative than the expected value because when the electrons are removed from the 4s-orbital. Zn acquires a stable $3d^{10}$ configuration state.

49. The halides of transition elements become more covalent with increasing oxidation state of the metal. Why?

Ans: As the positive charge of the ion increases or we can say that oxidation state of a transition element increases, its size decreases and as per Fajan's rule, more the charge on the metal ion, more is its tendency to form covalent compounds because positively charged cation attracts the electron cloud strongly towards itself.

50. While filling up of electrons in the atomic orbitals, the 4s-orbital is filled before the 3d-orbital but reverse happens during the ionization of the atom. Explain why?

Ans: As per (n+l) rule, 4s has lower energy than 3d-orbital.

$$3d - n + l = 3 + 2 = 5$$

$$4s - n + l = 4 + 0 = 4$$

So, 4s is filled first.

After filling of electrons, 4s-orbital moves beyond 3d-orbital and 4s electrons are loosely held by nucleus. Hence, electrons are removed first during the process of ionisation.

51. Reactivity of transition elements decreases almost regularly from Sc to Cu. Explain.

Ans: Ionisation enthalpies are the main factor that influence the reactivity of transition elements. Higher the ionisation enthalpy, lesser is the reacting of transition element.

When we move along the period from Sc to Cu, a regular increase in the ionisation enthalpy is observed which results in the almost regular decrease in the reactivity of elements.

Matching Type Questions

52. Match the catalysts given in Column I with the processes given in Column II.

Column I	Column II
(i) Ni in the presence of hydrogen	(a) Ziegler Natta catalyst
(ii) Cu_2Cl_2	(b) Contact process
(iii) V_2O_5	(c) Vegetable oil to ghee
(iv) Finely divided iron	(d) Sandmeyer reaction
(v) $\text{TiCl}_4 + \text{Al}(\text{CH}_3)_3$	(e) Haber's Process
	(f) Decomposition of KClO_3

Ans:

Column I	Column II
(i) Ni in the presence of hydrogen	(c) Vegetable oil to ghee
(ii) Cu_2Cl_2	(d) Sandmeyer reaction
(iii) V_2O_5	(b) Contact process
(iv) Finely divided iron	(e) Haber's Process
(v) $\text{TiCl}_4 + \text{Al}(\text{CH}_3)_3$	(a) Ziegler Natta catalyst

53. Match the compounds/elements given in Column I with uses given in Column II.

Column I (Compound/element)	Column II (Use)
(i) Lanthanoid oxide	(a) Production of iron alloy
(ii) Lanthanoid	(b) Television screen
(iii) Misch metal	(c) Petroleum cracking
(iv) Magnesium based alloy is constituent of	(d) Lanthanoid metal + iron
(v) Mixed oxides of lanthanoids are employed	(e) Bullets
	(f) In X-ray screen

Ans:

Column I (Compound/element)	Column II (Use)
(i) Lanthanoid oxide	(b) Television screen
(ii) Lanthanoid	(a) Production of iron alloy

(iii) Misch metal	(d) Lanthanoid metal + iron
(iv) Magnesium based alloy is constituent of	(e) Bullets
(v) Mixed oxides of lanthanoids are employed	(c) Petroleum cracking

54. Match the properties given in Column I with the metals given in Column II.

Column I (Property)	Column II (Metal)
(i) An element which can show +8 oxidation state	(a) Mn
(ii) 3d block element that can show upto +7 oxidation state	(b) Cr
(iii) 3d block element with highest melting point	(c) Os
	(d) Fe

Ans:

Column I (Property)	Column II (Metal)
(i) An element which can show +8 oxidation state	(c) Os
(ii) 3d block element that can show upto +7 oxidation state	(a) Mn
(iii) 3d block element with highest melting point	(b) Cr

55. Match the statements given in Column I with the oxidation states given in Column II.

Column I	Column II
(i) Oxidation state of Mn in MnO_2 is	(a) + 2
(ii) Most stable oxidation state of Mn is	(b) + 3
(iii) Most stable oxidation state of Mn in oxides is	(c) + 4
(iv) Characteristic oxidation state of lanthanoids is	(d) + 5

	(e) + 7
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Ans:

Column I	Column II
(i) Oxidation state of Mn in MnO_2 is	(c) + 4
(ii) Most stable oxidation state of Mn is	(a) + 2
(iii) Most stable oxidation state of Mn in oxides is	(e) + 7
(iv) Characteristic oxidation state of lanthanoids is	(b) + 3

56. Match the solutions given in Column I and the colours given in Column II.

Column I (Aqueous solution of salt)	Column II (Colour)
(i) $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$	(a) Green
(ii) $\text{NiCl}_2 \cdot 4\text{H}_2\text{O}$	(b) Light pink
(iii) $\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$	(c) Blue
(iv) $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$	(d) Pale green
(v) Cu_2Cl_2	(e) Pink
	(f) Colourless

Ans:

Column I (Aqueous solution of salt)	Column II (Colour)
(i) $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$	(d) Pale green
(ii) $\text{NiCl}_2 \cdot 4\text{H}_2\text{O}$	(a) Green
(iii) $\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$	(b) Light pink
(iv) $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$	(e) Pink
(v) Cu_2Cl_2	(f) Colourless

57. Match the property given in Column I with the element given in Column II.

Column I (Property)	Column II (Elements)
(i) Lanthanoid which shows +4 oxidation state	(a) Pm

(ii) Lanthanoid which can show +2 oxidation state	(b) Ce
(iii) Radioactive lanthanoid	(c) Lu
(iv) Lanthanoid which has $4f^7$ electronic configuration in +3 oxidation state	(d) Eu
(v) Lanthanoid which has $4f^{14}$ electronic configuration in +3 oxidation state	(e) Gd
	(f) Dy

Ans:

Column I (Property)	Column II (Elements)
(i) Lanthanoid which shows +4 oxidation state	(b) Ce
(ii) Lanthanoid which can show +2 oxidation state	(d) Eu
(iii) Radioactive lanthanoid	(a) Pm
(iv) Lanthanoid which has $4f^7$ electronic configuration in +3 oxidation state	(e) Gd
(v) Lanthanoid which has $4f^{14}$ electronic configuration in +3 oxidation state	(c) Lu

58. Match the properties given in Column I with the metals given in Column II.

Column I (Property)	Column II (Metal)
(i) Element with highest second ionisation enthalpy	(a) Co
(ii) Element with highest third ionisation Enthalpy	(b) Cr
(iii) M in $M(CO)_6$ is	(c) Cu
(iv) Element with highest heat of atomization	(d) Zn
	(e) Ni

Ans:

Column I (Property)	Column II (Metal)
(i) Element with highest second ionisation enthalpy	(c) Cu
(ii) Element with highest third ionisation Enthalpy	(d) Zn
(iii) M in $M(CO)_6$ is	(b) Cr
(iv) Element with highest heat of atomization	(e) Ni

Assertion and Reason Type Questions

59. Assertion (A): Cu^{2+} iodide is not known.

Reason (R): Cu^{2+} oxidises I to iodine.

Ans: All halogens combine with copper to form copper halides except iodine. The reason behind this is that Cu^{2+} oxidises iodide (-1) to iodine (0). Therefore, Cu^{2+} iodide does not exist. Both the statements assertion and reason are correct and reason is the correct explanation of assertion.

60. Assertion (A): Separation of Zr and Hf is difficult.

Reason (R): Because Zr and Hf lie in the same group of the periodic table.

Ans: Although Zr and Hf lie in same group but As an effect of lanthanide contraction, zirconium and hafnium have similar radius of 160pm and 159 pm respectively. Due to this similarity in their size, they show similar physical and chemical properties. As a result, Zr and Hf are difficult to separate. Both the statements assertion and reason are correct but reason is not the correct explanation of assertion.

61. Assertion (A): Actinoids form relatively less stable complexes as compared to lanthanoids.

Reason (R): Actinoids can utilize their 5d orbitals along with 6d orbitals in bonding but lanthanoids do not use their 4f orbital for bonding.

Ans: Actinoids are more reactive than lanthanoids as they involve their 5d-orbital in the bond formation process while due to effectively shielded and greater nuclear charge 4f-orbitals are not available for bonding. Thus, lanthanoids comparatively form less complexes than actinoids.

Assertion is not true but reason is true.

62. Assertion (A): Cu cannot liberate hydrogen from acids .

Reason (R): because it has positive electrode potential.

Ans: Cu lies below hydrogen in electrochemical series and has positive electrode potential. Hence cannot liberate hydrogen from acids.

Both the statements assertion and reason are correct and reason is the correct explanation of assertion.

63. Assertion (A): The highest oxidation state of osmium is +8.

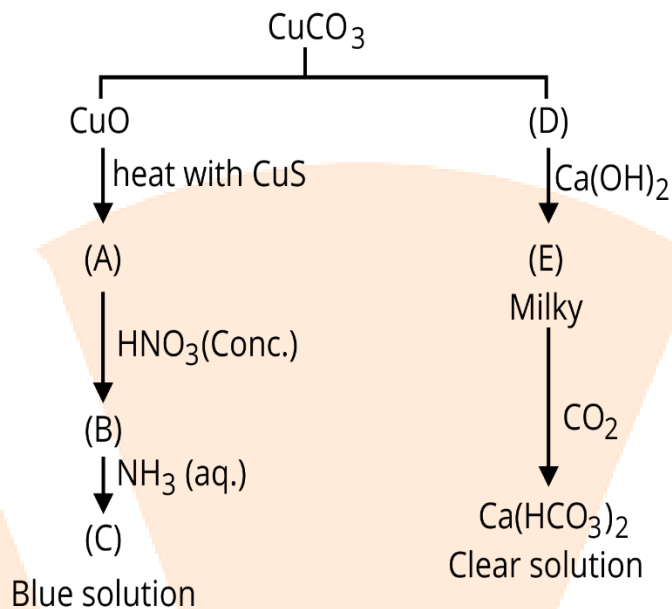
Reason (R): Osmium is a 5d-block element.

Ans: Osmium has the capacity to expand its octet by utilising all the electrons from its 6s and 5d orbitals this results in it attaining an oxidation state of +8.

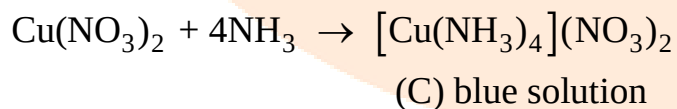
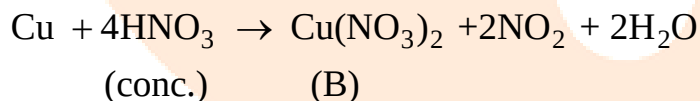
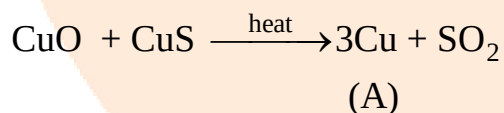
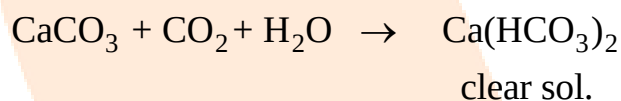
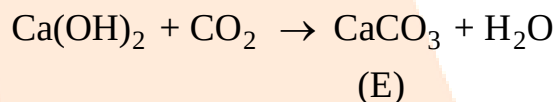
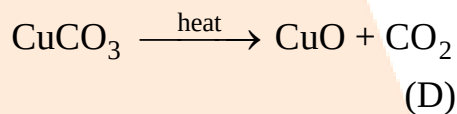
Both the statements assertion and reason are correct but reason is not the correct explanation of assertion. The correct option is B.

Long Answer Type Questions

64. Identify A to E and explain reaction involves:



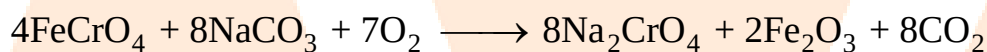
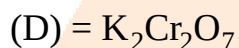
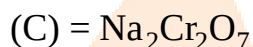
Ans: The following are the reactions:



65. When a chromite ore (A) is fused with sodium carbonate in free excess of air and the product is dissolved in water, a yellow solution of compound (B) is

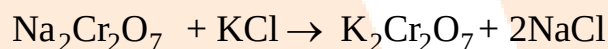
obtained. After treatment of this yellow solution with sulphuric acid, compound (C) can be crystallized from the solution. When compound (C) is treated with KCl, orange crystals of compound (D) crystallise out. Identify A to D and also explain the reactions.

Ans:



(A)

(B)

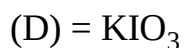


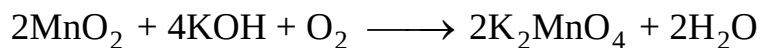
(C)

(D)

66. When an oxide of manganese (A) is fused with KOH in the presence of an oxidizing agent and dissolved in water, it gives a dark green solution of compound (B). Compound (B) disproportionate in neutral or acidic solution to give purple compound (C). An alkaline solution of compound (C) oxidises potassium iodide solution to a compound (D) and compound (A) is also formed. Identify compounds A to D and also explain the reactions involved.

Ans: (A) = MnO_2



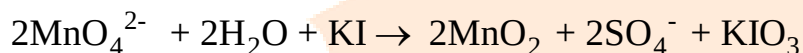


(A)

(B)



(C)



(A)

(D)

67. On the basis of lanthanoid contraction, explain the following:

(a) Nature of bonding in La_2O_3 and Lu_2O_3 .

Ans: As per fajan's rule, smaller the size of ion, greater is its tendency to make covalent bonds as the positively charged cation strongly attract the negatively charged electron cloud.

On moving from La to Lu in the lanthanoid series, the atomic size decreases so the covalent character increases. Hence, La_2O_3 is ionic and Lu_2O_3 is covalent.

(b) Trends in the stability of oxo-salts of lanthanoids from La to Lu.

Ans: The stability of Oxo salts is directly proportional to the size of atom, as we move from La to Lu, the size of the atom decreases and hence the stability of oxo salts also decreases.

(c) Stability of the complexes of lanthanoids.

Ans: As we move along the lanthanoid series, the atomic size decreases. As a result, the charge/size ratio increases and the stability of the complexes also increases.

(d) Radii of 4d- and 5d-block elements.

Ans: As an effect of lanthanoid contraction, zirconium and hafnium have similar radius of 160pm and 159 pm respectively. Due to this similarity in their size, they show similar physical and chemical properties. Lanthanoid contraction: Because the elements in Row 3 have 4f electrons. These electrons do not shield good, causing a

greater nuclear charge. This greater nuclear charge has a greater pull on the electrons and result in the decrease in their size and atomic radii.

(e) Trends in acidic character of lanthanoid oxides.

Ans: Acidic or basic character of oxides is related to the nature of oxides. As we move along the lanthanoid series from La to Lu, the covalent character increases.

Therefore, their basic character decreases or acidic character increases.

68. (A) Answer the following questions:

(i) Which element of the first transition series has highest second ionization enthalpy?

Ans: Out of all the elements of first transition series copper has the highest second ionisation enthalpy.

Electronic configuration of Copper is: $3d^{10}4s^1$

After the Loss of first electron from the 4s copper acquires $3d^{10}$ configuration which is stable. Therefore, removal of second electron from the field 3-D orbital is very difficult and requires high amount of energy.

(ii) Which element of the first transition series has highest third ionization enthalpy?

Ans: Among the elements of first transition series zinc has the highest third ionisation enthalpy. Electronic configuration of zinc is: $3d^{10}4s^2$

After the loss of two electrons from 4s orbital, Z and +2 Ion acquires $3d^{10}$ fully filled configuration which is highly stable therefore removal of third electron from $3d^{10}$ orbital will be more difficult and requires a large amount of energy.

(iii) Which element of the first transition series has lowest enthalpy of atomization?

Ans: Zinc is the element of the first transition series that has the lowest enthalpy of atomisation. The electronic configuration of zinc is: $3d^{10}4s^2$. This is because it has filled 3d-subshell and filled 4s-sub shell so no unpaired electron is available for metallic bonding.

(b) Identify the metal and justify your answer:

(i) Carbonyl $M(CO)_5$

Ans: $Fe(CO)_5$ as per EAN Rule.

EAN Rule: Effective atomic number (EAN), number that represents the total number of electrons surrounding the nucleus of a metal atom in a metal complex.

(ii) MO_3F

Ans: The M in the given complex is in +7 O.S.

Manganese is the only element in the first transition series that shows +7 oxidation state, therefore this compound is MnO_3F .

69. Mention the type of compounds formed when small atoms like H, C and N get trapped inside the crystal lattice of transition metals. Also give physical and chemical characteristics of these compounds.

Ans: Interstitial compounds are formed when small atoms like H, C and N get trapped inside the crystal lattice of transition metals.

Characteristics:

They are hard and rigid in nature having high melting points.

Like pure metals only, they are conductors. They can gain chemical inertness.

70. (A) Transition metals can act as catalysts because these can change their oxidation state.

How does Fe(III) catalyse the reaction between iodide and persulphate ions?



Role of Fe ions:



(B) Mention any three processes where transition metals act as catalysts.

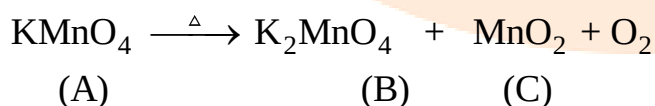
Ans: 1. Fine powdered state of Nickel is used in the hydrogenation of oils into fats.

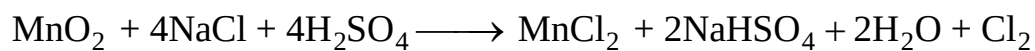
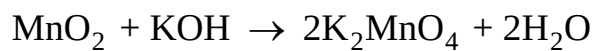
2. Iron is used in the formation of ammonia in Haber's process.

3. In contact process in manufacture of H_2SO_4 , vanadium is used in its oxide form V_2O_5 .

71. A violet compound of manganese (A) decomposes on heating to liberate oxygen and compounds (B) and (C) of manganese are formed. Compound (C) reacts with KOH in the presence of potassium nitrate to give compound (B). On heating compound (C) with conc. H_2SO_4 and NaCl, chlorine gas is liberated and a compound (D) of manganese along with other products is formed. Identify compounds A to D and also explain the reactions involved.

Ans:





(D)

(A) = KMnO_4

(B) = K_2MnO_4

(C) = MnO_2

(D) = MnCl_2